

ଉତ୍ତରମାଳା

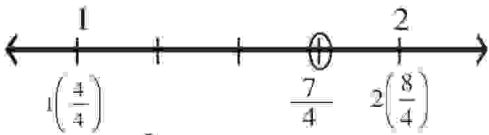
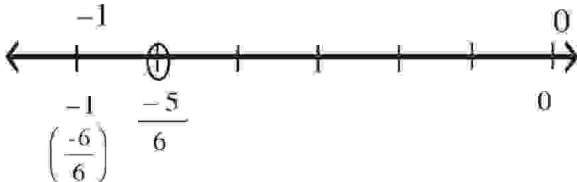
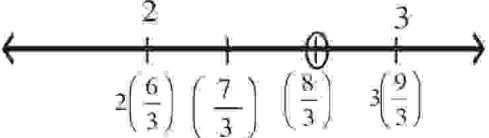
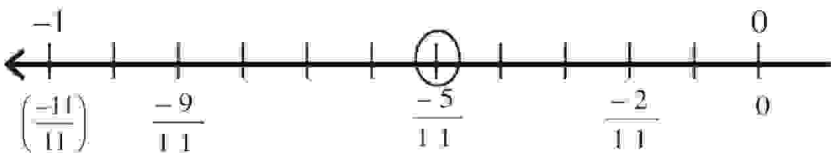
ଅନୁଶୀଳନୀ - 1

- ଠିକ୍ ଉକ୍ତି : (i), (iv), (vi), (ix) ଏବଂ (xi); ଅବଶିଷ୍ଟ ଭୁଲ୍ ଉକ୍ତି ।
- (i) $\in$ (ii) $\subset$ (iii) $\subset$ ବା $=$ (iv) $\notin$ (v) $\subset$ (vi) $\supset$, 3. (i) {2, 3, 4, 5, 6, 7, 8, 9} (ii) {2, 4, 6, 8} (iii) {2} (iv) {2, 4, 6, 8}, (v) {-5, -4, -3, -2, -1, 0, 1, 2, 3}, (vi) {ସୋମ, ମଙ୍ଗଳ, ବୁଧ, ଗୁରୁ, ଶୁକ୍ର, ଶନି, ରବି}, (vii) { }, (viii) {8, 16};
- (i) $\{x \mid x \text{ ଏକ ଅଯୁଗ୍ମ ଗଣନ ସଂଖ୍ୟା ଓ } x < 12\}$, (ii) $\{x \mid x \text{ ଇଂରାଜୀ ବର୍ଣ୍ଣମାଳାର ଗୋଟିଏ ସ୍ଵରବର୍ଣ୍ଣ}\}$ (iii) $\{x \mid x \text{ ଏକ ପୂର୍ଣ୍ଣ ସଂଖ୍ୟା, } -2 \leq x \leq 2\}$, (iv) $\{x \mid x \text{ ଏକ ମୌଳିକ ସଂଖ୍ୟା, } x < 14\}$ (v) $\{2n \mid n \in \mathbb{N}\}$ (vi) $\{3n \mid n \in \mathbb{N} \text{ ଏବଂ } n \leq 5\}$, (vii) $\{x \mid x = 5^n, n \in \mathbb{N} \text{ ଏବଂ } n \leq 4\}$ (viii) $\{x \mid x \text{ ଇଂରାଜୀ ବର୍ଣ୍ଣମାଳାର ଏକ ଅକ୍ଷର}\}$, (ix) $\{x \mid x = 2^n, n \in \mathbb{N}\}$
- (i) {m, a, t, h, e, i, c, s}, (ii) {a, r, i, t, h, m, e, c}, (iii) {p, r, o, g, a, m, e}, (iv) {c, o, m, i, t, e}
- $A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$, $A \cap B = \{2, 4, 6\}$, 7. $A \cup B = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A \cap B = \{5, 6\}$, 8. (i) {1, 2, 3, 4, 5}, (ii) {2, 4}, (iii) {2}, (iv) {1, 2, 3, 4, 6}, (v) {2, 3, 4, 5, 6}, (vi) {2, 3}, 9. (i) $A = \{1, 3, 4, 5\}$, $B = \{2, 4, 5, 6, 7\}$, (ii) {4, 5}, (iii) {1, 2, 3, 4, 5, 6, 7}, (iv) {1, 3}, (v) {2, 6, 7}, 10. (i) $A = \{1, 2, 3, 4, 5, 6, 7\}$, $B = \{3, 4, 5\}$, (ii) $A \cap B = \{3, 4, 5\}$, (iii) $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, (iv) $A \cup \phi = \{1, 2, 3, 4, 5, 6, 7\}$, (v) $A \cap \phi = \{ } \text{ ବା } \phi$, 11. (a) {a, b}, {e, f}, (b) {a, b, e, f}, (c) ϕ ବା { }

ଅନୁଶୀଳନୀ - 2 (a)

- (i) $-\frac{2}{8}$, (ii) $\frac{5}{9}$, (iii) $-\frac{6}{5}$, (iv) $\frac{2}{9}$, (v) $\frac{19}{6}$, 2. (i) $-\frac{1}{13}$, (ii) $\frac{19}{13}$, (iii) 5, (iv) $\frac{56}{15}$, (v) $\frac{5}{2}$ (vi) -1,
- (i) (ଗୁଣନାତ୍ମକ) ଅଭେଦ ନିୟମ, (ii) ଗୁଣନର କ୍ରମ ବିନିମୟ ନିୟମ, (iii) (ଗୁଣନାତ୍ମକ) ବିଲୋମୀ ନିୟମ, (iv) ଗୁଣନର ସହଯୋଗୀ ନିୟମ, 4. $\frac{-96}{91}$, 5. ବ୍ୟୁତ୍କ୍ରମ, 6. ବ୍ୟୁତ୍କ୍ରମ, 7. (i) 0, (ii) 1 ଓ -1, (iii) 0
- (i) ନାହିଁ, (ii) 1 ଓ -1 (iii) $-\frac{1}{5}$ (iv) x (v) ପରିମେୟ ସଂଖ୍ୟା (vi) ରଶୀମୂଳ ପରିମେୟ ସଂଖ୍ୟା

ଅନୁଶୀଳନୀ - 2 (b)

- (i)  (ii)  (iii) 
- 
- (i) $1, \frac{1}{2}, 0, -1, -\frac{1}{2}$, (ii) $-\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}$ (ଅନ୍ୟ ଉତ୍ତର ମଧ୍ୟ ସମ୍ଭବ), 6. (i) $\frac{5}{7}$ (ii) $\frac{7}{9}$ (iii) $\frac{3}{7}$

ଅନୁଶୀଳନୀ - 2 (c)

1. (a) $1234 \times 9 + 4 = 11110$

$12345 \times 9 + 5 = 111110$

(c) $1537 \times 10001 = 15371537$

$24631 \times 100001 = 2463124631$

(e)
$$\begin{array}{cccccc} 1 & 5 & 10 & 10 & 5 & 1 \\ 1 & 6 & 15 & 20 & 15 & 6 & 1 \end{array}$$

(b) $1234 \times 8 + 4 = 9876$

$12345 \times 8 + 5 = 98765$

(d) $16 + 17 + 18 + 19 + 20 = 21 + 22 + 23 + 24$

$25 + 26 + 27 + 28 + 29 + 30 = 31 + 32 + 33 + 34 + 35$

(f) $7^2 - 6^2 = 7 + 6 = 13$

$8^2 - 7^2 = 8 + 7 = 15$

2. (i)

	29	
83	23	17
	71	

(ii)

		23		
		37		
13	47	17	73	11
		41		
		43		

3. (i) $x = 2, y = 4$, (ii) $x = 2, y = 3, z = 6$, (iii) $A = 3, C = 7$; (iv) $A = 1, B = 0, C = 8, D = 9$,
(v) $A = 3, B = 7, P = 11$, (vi) $A = 7, B = 3, P = 9$, (vii) ଓ (viii) A, B, C ର ମାନ ଯେକୌଣସି ଏକ
ଅଙ୍କ ବିଶିଷ୍ଟ ସଂଖ୍ୟା ହୋଇପାରେ ।

4. (a) 24, 210, 86, (b) 5 ଦ୍ଵାରା ବିଭାଜ୍ୟ ସଂଖ୍ୟା : 105, 420, 235, 930, 715; 5 ଓ 2 ଉଭୟ ଦ୍ଵାରା ବିଭାଜ୍ୟ
ସଂଖ୍ୟା : 420, 930, (c) 3 ଦ୍ଵାରା ବିଭାଜ୍ୟ ସଂଖ୍ୟା : 78, 504, 216, 774, 804; (d) 501, 213, 102, 462
ଓ 2 ଉଭୟ ଦ୍ଵାରା ବିଭାଜ୍ୟ ସଂଖ୍ୟା : 420, 930, 5. (a) 0.6 (b) 1, 4 (c) 2.2 (d) 1.4 (e) 2.5:

6. (i), (iv), (v) : ଠିକ୍ ଉଚ୍ଛି, 7. (ii), (iv), (v) : ଠିକ୍ ଉଚ୍ଛି

ଅନୁଶୀଳନୀ - 3 (a)

1. (i) 2, $5x$ (ii) 5, $12x$, (iii) $-6, 4, -2x$ (iv) $-2, -3, -5x$ (v) 1, $-2, -x$

2. (i) $7x$, (ii) $-x$, (iii) $-5x^3$, (iv) $-3x^2$ (v) $4x - 4$ (vi) $3x^2 + 2$ (vii) $x^2 + x$ (viii) $x^2 + 3x + 4$

3. (i) $5x$, (ii) $7x$, (iii) $4x$, (iv) $3x$, (v) $5x, 7x$ (vi) $2x + 5y, 5y$.

4. (i) $10x$, (ii) $9x^2$ (iii) $9x^3$ (iv) $4x^2 + 5x$ (v) $x^3 - x^2 + x + 7$ (vi) $2x^2 + 2x$ (vii) $2x^2$, (viii) $2x^2 + 6x - 4$,
(ix) $6x^2 - x + 4$ (x) 0 :

ଅନୁଶୀଳନୀ - 3 (b)

1. (i) $-3x, 5, -3, 2x$ (ii) $2x, 3, 2, 5x$ (iii) $-3x, -2, -3, -5x$, (iv) $-3 + 2x, 2x - 1, 5x$

(v) $3x - 2, -4 - 2, 4x, -6$; 2. (i) $3x$ (ii) $8x$ (iii) $-5x$ (iv) $2x$ (v) $-2x$ (vi) $2 - x^2 - x$ (vii) $x^3 - 4x - 6$;

3. (i) $2x$ (ii) $4x^2$ (iii) $x^2 - 4x$ (iv) $2x^2 + 4$ (v) $x^2 - 10x$ (vi) $6x + 8$ (vii) $x^3 - 30x - 8$

ଅନୁଶୀଳନୀ - 3 (c)

1. (i) $15x$, (ii) $6x^4$, (iii) 0, (iv) $3x^3$, 3. (i) -7 (ii) 1 (iii) $-x, -6$ (iv) $-3x^2, 4$

4. (i) $x^2 - 1$ (ii) $x^3 - 1$ (iii) $x^3 + 1$ (iv) $2x^2 - 3x - 2$ (v) $2x^3 - x^2 + 4x + 15$ (vi) $-x^3 + 2x^2 + 17x + 6$.
(vii) $x^4 - 1$ (viii) $2x^4 - x^3 + 3x^2 - x + 1$, (ix) $x^4 + x^3 - x - 1$

ଅନୁଶୀଳନୀ - 3 (d)

1. (i) $4x$, (ii) $6x$, (iii) $-4x^2$, (iv) $-5x$, 2. (i) $2x$ (ii) $-2x$ (iii) $-2x$, (iv) $2x$; 3. (i) $7x^2$ (ii) $-7x$ (iii) $-3x$,

(iv) 7, (v) -7 ; 4. (i) $3x^3 + 2$ (ii) $4x^2 - 3$ (iii) $6x^2 - 2x + 3$ (iv) $4x + 3$, (v) $6x + 5$, (vi) $-12x + 11$

ଅନୁଶୀଳନୀ - 3 (e)

1. (i) $6x^2 + 3x$, (ii) x , (iii) $3x^2 + 2$, (iv) 1, (v) $12x^2 + 8x + 2$; 2. (i) $x - 7$, (ii) $x - 4$, (iii) $x + 5$,
(iv) $x - 1$, (v) $x^2 - x + 1$, (vi) $x^2 + x + 1$ (vii) $x^2 - x + 1$, (viii) $-x^2 + 5x - 6$, (ix) $x^2 - x - 2$,

(x) $2x^2 + 3x + 2$, 3. (i) $(x+14)$, 42 (ii) $(x-11)$, 19 (iii) $(2x-2)$, -9 (iv) $(9x^2 + 2)$, 3,
(v) $4x^2 - 2x + 1$, -2 (vi) $-x^2 + x - 1$, -2, 4. (i) -14, (ii) 2, (iii) -2

ଅନୁଶୀଳନୀ - 3 (f)

1. (i) 4, (ii) $2y$, (iii) $-4y$, (iv) $4xy$, (v) $(a-b)$; 2. (i) $b^2 + 2bc + c^2$, (ii) $16+8b+b^2$, (iii) $r^2-20r+100$, (iv) $9n^2+12n+4$, (v) $4m^2+4mn+n^2$, (vi) $49p^2-14pq+q^2$, (vii) $4x^2+12xy+9y^2$, (viii) $4m^2+9n^2+p^2-12mn+6np-4mp$, (ix) $x^2+y^2+16z^2-2xy-8yz+8xz$, (x) $a^2+4b^2+9c^2+4ab+12bc+6ac$; 3. (i) 10404, (ii) 92416, (iii) 1006009, (iv) 16008001; 4. (i) 9801, (ii) 996004, (iii) 89991, (iv) 6396, (v) 79.21, (vi) 9.975, (vii) 200, (viii) 0.08, (ix) 1800; 5. (i) 10712, (ii) 26.52, (iii) 10094, (iv) 95.06; 6. (i) x^2+6x+9 , (ii) $4y^2+20y+25$, (iii) $4a^2-28a+49$, (iv) $1.21m^2-0.16$, (v) b^2-a^2 , (vi) $36x^2-49$, (vii) p^2-25 , (viii) $9y^2-4x^2$, (ix) x^2-1 , (x) $16y^2-81$; 7. (i) $x^2+10x+21$, (ii) $16x^2+24x+5$, (iii) $16x^2-24x+5$, (iv) $16x^2+16x-5$, (v) $4a^2+28a+45$, (vi) $xyz-6xyz+8$; 8. (i) $2a^2+2b^2$, (ii) $40x$, (iii) $98m^2+128n^2$, (iv) $41m^2+80mn+41n^2$, (v) $4p^2-4q$, (vi) $a^2b^2+b^2c^2$, (vii) m^2+n^2 , (viii) $2a^2+2b^2+2c^2-4ca$, (ix) $8a^2+10b^2+26c^2-16ab-4bc+16ac$, (x) $8x^2+12y^2-20xy-12yz+8xz$; 9. (i) $(2x+3y)^2$, (ii) $(8m-3n)^2$, (iii) $(2x-1)^2$, (iv) $(x+2y+z)^2$, (v) $(2x-y-z)^2$, (vi) $(3x-2y+z)^2$

ଅନୁଶୀଳନୀ - 4 (a)

1. $12(x+3)$, 2. $4(2a+b)$, 3. $11(2y-3z)$, 4. $7pq(2+5r)$, 5. $5a(2ab+1)$, 6. $5abc(3a-2b)$, 7. $2a(4a^2+2a+1)$, 8. $5a^3b^3c^3(6+5a^2c^3-3a^3b^3c^3)$, 9. $10(2x+5)$, 10. $(2x+3y)(5a-2b)$, 11. $4(5x+9y)(10x+18y+3)$, 12. $3a(6a-5b)(3-4a)$, 13. $(x-2y)(5x-10y+3)$, 14. $2(a+2b)(3-2a-4b)$, 15. $(a-1)(a+b)$, 16. $(x-y)(x-y+1)$, 17. $(x-y)(a+2b+c)$, 18. $(b-c)(a+b+c)$, 19. $x^2(a-2b)(x+1)$, 20. $2(x+y)(4b-3a)$, 21. $5(a+b)(x-y)$, 22. $(x+y)(a^2+b^2+x^2)$

ଅନୁଶୀଳନୀ - 4 (b)

1. $(x+y)(x+8)$, 2. $(q+r)(p+q)$, 3. $(a+d)(b+c)$, 4. $(p+r)(q+r)$, 5. $(5y-2)(3x+1)$, 6. $(a+b)(x-y)$, 7. $(5p+3)(3q+5)$, 8. $(a+3b)(2-3a-9b)$, 9. $(a+2)(a+b)$, 10. $(x-z)(x+y)$, 11. $(a-b)(a-c)$, 12. $(2p-q)(p-r)$, 13. $(x-3)(x+2)$, 14. $(2x-5)(x+2)$, 15. $(x+1)(x-y^2)$, 16. $(m-n^2)(m-l)$, 17. $(x-2y)(x^2+3y^2)$, 18. $(6a-b)(b+2c)$, 19. $(x-11y)(x-1)$, 20. $(3a+4b)(x-2y)$

ଅନୁଶୀଳନୀ - 4 (c)

1. (i) $(a+3)(a+5)$, (ii) $(x+2)(x+3)$, (iii) $(x+6)(x+1)$, (iv) $(x+6)(x+2)$, (v) $(x+3)(x+8)$, (vi) $(x+1)(x+1)$; 2. (i) $(p-4)(p-6)$, (ii) $(x-2)(x-6)$, (iii) $(x-2)(x-5)$, (iv) $(x-2)(x-7)$, (v) $(x+7)(x-3)$, (vi) $(x-2)(x-1)$; 3. (i) $(a+1)(a-5)$, (ii) $(x+3)(x-14)$, (iii) $(x+3)(x-7)$, (iv) $(x+9)(x-10)$, (v) $(x+7)(x-9)$, (vi) $(x-2)(x+1)$; 4. (i) $(a+7)(a+11)$, (ii) $(a-6)(a-2)$, (iii) $(x+2)(x-4)$, 5. $(a-9)(a+6)$, 6. $(x-2y-3)(x-2y-2)$

ଅନୁଶୀଳନୀ - 4 (d)

1. (i) $(2x+1)(2x+1)$, (ii) $(3b+2c)(3b+2c)$, (iii) $(4a+5b)(4a+5b)$, (iv) $(7x+8y)(7x+8y)$, (v) $(a^2+3b^2)(a^2+3b^2)$; 2. (i) $(3x-1)(3x-1)$, (ii) $(4x-5y)(4x-5y)$, (iii) $(7a-9b)(7a-9b)$, (iv) $(8a-1)(8a-1)$, (v) $(10a^2-b)(10a^2-b)$; 3. (i) $(4x+3y+5z)(4x+3y+5z)$, (ii) $(7x+5y+z)(7x+5y+z)$, (iii) $(2a+3b-c)(2a+3b-c)$, (iv) $(10a-9b-7c)(10a-9b-7c)$, (v) $(x^2-y-z)(x^2-y-z)$; 4. (i) $(4a+3b)(4a-3b)$, (ii) $(5a+6b)(5a-6b)$, (iii) $(9a+10b)(9a-10b)$, (iv) $(4a+7b)(4a-7b)$, (v) $(12a+15b)(12a-15b)$ or $9(4a+5b)(4a-5b)$, (vi) $(16a+17b)(16a-17b)$,

(vii) $(20a + 15b)(20a - 15b)$ or $25(4a + 3b)(4a - 3b)$, (viii) $(21a + 30b)(21a - 30b)$ or $9(7a + 10b)(7a - 10b)$, (ix) $(11a + 17b)(11a - 7b)$, (x) $(9a + 19b)(9a - 19b)$, (xi) $(a + b + c)(a + b - c)$, (xii) $(a + b - c)(a - b + c)$.

5. (i) $(a^2 + 1 + a)(a^2 + 1 - a)$, (ii) $(x^2 + x + 1)(x^2 - x + 1)$, (iii) $(x^2 + 36y^2 + 6xy)(x^2 + 36y^2 - 6xy)$, (iv) $(x^2 + 3xy + 9y^2)(x^2 - 3xy + 9y^2)$, (v) $(x^2 + 4x + 16)(x^2 - 4x + 16)$.

6. (i) $(a + 3 + b)(a + 3 - b)$, (ii) $(a - 2 + c)(a - 2 - c)$, (iii) $(2a - 1 + 3b)(2a - 1 - 3b)$, (iv) $(a - 3b + 4c)(a - 3b - 4c)$, (v) $(4a - 3b + 5c)(4a - 3b - 5c)$.

7. (i) $(x + 13)(x - 15)$, (ii) $(x + 21)(x - 17)$, (iii) $(x + 14)(x - 8)$, (iv) $(x + 31)(x - 29)$, (v) $(x - 27)(x + 23)$, (vi) $(x - 19)(x + 9)$, (vii) $(x - 33)(x + 27)$, (viii) $(x + 16)(x - 12)$.

ଅନୁଶୀଳନ - 5 (a)

1. (i) 2^4 , (ii) $(-2)^5$, (iii) $\left(\frac{3}{4}\right)^3$, (iv) $\left(-\frac{1}{7}\right)^4$, (v) $\left(\frac{5}{3}\right)^3$

(vi) y^5 (vii) $(-p)^3$, (viii) $(a - b)^4$, (ix) $(a + b)^3$, (x) $\left(\frac{a}{b}\right)^5$

କ୍ରମିକ ନମ୍ବର	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
ଆଧାର	1	-1	-1	9	-2	$\frac{1}{2}$	$\frac{2}{3}$	10	10	-10
ଘାତ	15	11	18	5	5	6	5	4	7	5
ମାନ	1	-1	1	59049	-32	$\frac{1}{64}$	$\frac{32}{243}$	10000	10000000	-100000

3.	ସମ୍ଭା	୧ମ	୨ୟ	୩ୟ	୪ର୍ଥ	୫ମ	୬ଷ୍ଠ	୭ମ	୮ମ
	ଉତ୍ତର	64	729	2	5	4	5	$-\frac{1}{128}$	$-\frac{1}{3}$

4. (i) 10000, (ii) ଚତୁର୍ଥ, (iii) ତୃତୀୟ, (iv) $-\frac{3}{2}$ (v) $\frac{1}{25}$ 5. (i) ତୃତୀୟ, (ii) 25, (iii) 64

ଅନୁଶୀଳନ - 5 (b)

1. (i) 3^{10} , (ii) $\left(\frac{1}{2}\right)^{11}$ (iii) $\left(\frac{2}{3}\right)^{10}$, (iv) $(-4)^9$, (v) $\frac{3}{2}$, (vi) $(4)^9$, (vii) $(3)^{18}$, (viii) $(2)^{17}$, (ix) $(-7)^{13}$,

(x) $(2)^7$, (xi) $(5)^{12}$, (xii) $(-2)^{12}$ ବା 2^{12} , (xiii) $\left(\frac{7}{3}\right)^4$, (xiv) $\left(\frac{3}{4}\right)^9$ (xv) $\left(\frac{a}{b}\right)^{10}$, (xvi) $\left(\frac{-a}{b}\right)^7$

2. (i) 9, (ii) 972, (iii) 8, (iv) 64, (v) $\frac{6651}{256}$; 3. (i) 512, (ii) a^8b^7 , (iii) a^7b^4 , (iv) 1, (v) 1

4. (i) 2^{18} , (ii) 3^{14} , (iii) $5^{(3m-3)}$, (iv) $(-2)^{33}$, 5. (i) F, (ii) T, (iii) F, (iv) T, (v) T, (vi) F, (vii) T, (viii) F, (ix) F, (x) T; 6. (i) (iv) (v) (vii) (viii) ଓ (x)

ଅନୁଶୀଳନ - 5 (c)

1. (i) $\frac{1}{4}$, (ii) $\frac{1}{16}$ (iii) $\frac{1}{27}$, (iv) $\frac{1}{243}$, (v) $\frac{1}{10000}$, (vi) $\frac{1}{125}$, (vii) $\frac{1}{8000}$, (viii) $\frac{1}{125000}$, (ix) $\frac{1}{100}$,

(x) $\frac{1}{100000}$, (xi) -1, (xii) -1; 2. (i) 9, (ii) $\frac{125}{8}$, (iii) 10000, (iv) 0.008 ବା $\frac{1}{125}$, (v) $\frac{125}{27}$, (vi) $\frac{1000}{27}$,

(vii) -1, (viii) 1, 3. (i) $\left(\frac{1}{3}\right)^{-6}$, (ii) $\left(\frac{1}{6}\right)^{-3}$, (iii) $\left(-\frac{1}{6}\right)^{-3}$, (iv) $\left(\frac{1}{5}\right)^{-4}$, (v) $\left(\frac{1}{7}\right)^{-3}$, (vi) 8^{-3} (vii) $\left(\frac{3}{2}\right)^{-6}$

ଅନୁଶୀଳନ - 5 (d)

1. (i) 16, (ii) 32 (iii) 3125, (iv) $\frac{3}{5}$, (v) 36, (vi) 243, 2. (i) 2, (ii) 4, (iii) 81, (iv) $\frac{1}{2}$, (v) 625, (vi) $\frac{7}{2}$,

3. (i) 1, (ii) 1, 4. (i) $a - b$, (ii) $x - y$

ଅନୁଶୀଳନୀ - 6 (a)

1. 729, 1369, 2116, 13924, 50625;
3. 28, 278, 314, 23872..... ବର୍ଗ ଯୁଗ୍ମ ସଂଖ୍ୟା
113, 4315, ବର୍ଗ ଅଯୁଗ୍ମ ସଂଖ୍ୟା
5. $10^2 - 9^2$, $14^2 - 13^2$, $16^2 - 15^2$, $21^2 - 20^2$, $27^2 - 26^2$, 6. (7, 24, 25), (11, 60, 61), (15, 112, 113),
(12, 35, 37), (16, 63, 65), 8. 35, 49, 223, 341 9. (a), (b), (c), (d), (f), (g) - ଭୁଲ୍ (c), (e) - ଠିକ୍

ଅନୁଶୀଳନୀ - 6 (b)

1. 2025, 3025, 7225, 11025, 24025, 65025, 2. 729, 1369, 2116, 6084, 9604,
3. 361, 10404, 11449, 4. 8649, 9025, 9604, 5. 2601, 2916, 3136, 3364, 3481, 6. 1225, 5626,
9025, 13225, 42025; 7. 0.0144, 1.2321, 0.000009, 8. 121, 65.61, 0.36

ଅନୁଶୀଳନୀ - 6 (c)

1. (a) 0.6, (b) 1.1 (c) $1\frac{1}{3}$ (d) 0.03, (e) $2\frac{1}{2}$, 2. 17, 19, 28, 2.5, 3.6, 4.4, 3.2, 3. 305, 316, 329, 273,
1502, 1371, 231, 4. 7.29, 6.03, 2.098, 0.99, 2.34, 5. (i) 2.236, (ii) 2.645, (iii) 3.162, (iv) 1.581,
(v) 1.897, 6. 1.117, 1.666, 2.015, 1.811, 2.136, 7. (i) 3.535, (ii) 1.539, (iii) 3.732, (iv) 0.102, (v) 9.898

ଅନୁଶୀଳନୀ - 6 (d)

1. 961 ଏବଂ 1024, 2. 50, 3. 35, 4. 3, 5. 48, 6. 144 ମିଟର, 7. 150 ମିଟର, 8. 580 ଟଙ୍କା, 9. 120 ଟଙ୍କା,
10. 40, 11. 25, 12. 46.24 ମି.

ଅନୁଶୀଳନୀ - 6 (e)

1. 1331, 1728, 2197, 2744, 3375, 4096, 4913, 5832, 6859, 8000
2. (i) 12, (ii) 55, (iii) 60, (iv) 3, (v) 3; 3. 216, 4. 5, 5. 5; 6. 3375 ଘ. ଘେ.ମି., 7. 8; 8. 43200 ଟଙ୍କା,

ଅନୁଶୀଳନୀ - 6 (f)

1. (i) 7, (ii) 10, (iii) 42, (iv) 54, (v) 200; 2. 1, 14; 3. 2.14; 4. 64, ବ.ମି., 5. 5, 25

ଅନୁଶୀଳନୀ - 6 (g)

1. -1, -5, -18, -26, -140; 2. 8; 3. -72; 4. -56; 5. 75; 6. 225; 7. -77, 8. 60; 9. 14; 10. -9;
11. 12; 12. ପୂର୍ଣ୍ଣ ଘନରାଶି, -64, -1728, -2197 ଏବଂ ଘନମୂଳ -4, -12, -13
13. (i) -30, (ii) -72, (iii) -300, (iv) -80

ଅନୁଶୀଳନୀ - 6 (h)

1. (i) $\frac{343}{729}$, (ii) $\frac{-512}{1331}$, (iii) $\frac{1728}{343}$, (iv) $\frac{-2197}{512}$, (v) $17\frac{72}{125}$, (vi) $34\frac{21}{64}$, (vii) $-\frac{125}{27}$, (viii) 0.008
(ix) 2.197, (x) 0.000027; 2. (i) $\frac{2}{5}$, (ii) $-\frac{4}{11}$, (iii) $\frac{-3}{16}$, (iv) $\frac{13}{21}$, (v) 0.1 (vi) 0.2, (vii) 1.2, (viii) 0.05;
3. (i), (iii), (iv) (vi)

ଅନୁଶୀଳନୀ - 7 (a)

1. (i) 9, (ii) 7, (iii) 4, (iv) 21, (v) 5, (vi) 5, (vii) 50, (viii) 2.4, (ix) -11, (x) 2.1
2. (i) 4, (ii) 16, (iii) -5, (iv) 17, (v) 9, (vi) 2, (vii) $\frac{9}{25}$, (viii) 12 (ix) 2, (x) -10
3. (i) 10, (ii) -1, (iii) -26, (iv) -12 $\frac{2}{3}$, (v) -2 4. (i) $12\frac{1}{6}$, (ii) $\frac{2}{5}$, (iii) -26, (iv) $2\frac{1}{4}$, (v) -10

ଅନୁଶୀଳନୀ 7 (b)

1. 80, 2. 72, 3. 8, 4. 11, 5. 15, 16, 6. 14, 7. ହମିଦ୍‌ର ଟ.200 ଓ ରହିମ୍‌ର ଟ.150ଙ୍କା ।
8. 36, 9. 37, 10. 150, 11. 20, 50; 12. ପୁଅ-21 ଓ ଝିଅ-28, 13. 40° , 50° , 14. 5 ଟଙ୍କା 50 ଟି
ଓ 10 ଟଙ୍କା 25ଟି, 15. ପ୍ରଶ୍ନ 25ମି. ଓ ଚୈତ୍ୟ 50ମି. 16. $\frac{9}{12}$ 17. 70° 18. 4 କି.ମି.

ଅନୁଶୀଳନୀ - 7 (c)

1. (i) 0 ଓ 3, (ii) $\frac{5}{2}$ ଓ $-\frac{5}{2}$, (iii) -2 ଓ 2, (iv) $-\frac{4}{3}$ ଓ $\frac{4}{3}$, (v) $-\frac{5}{2}$ ଓ 0 (vi) 0 ଓ $\frac{b}{a}$
 (vii) -9 ଓ 9, (viii) -27 ଓ 27; 2. (i) 3 ଓ -1, (ii) 5 ଓ -1, (iii) 5 ଓ -4, (iv) -3 ଓ -4
 (v) -7 ଓ 5, (vi) 5 ଓ 1, (vii) -1 ଓ $\frac{3}{2}$, (viii) 1 ଓ $-\frac{5}{3}$, (ix) a ଓ b, (x) -a ଓ b

ଅନୁଶୀଳନୀ - 8 (a)

1. 500ଟଙ୍କା 2. 16% 3. 25% 4. $56\frac{1}{4}\%$ 5. $33\frac{1}{3}\%$ 6. $5\frac{5}{8}$ ଟଙ୍କା 7. 650 ଟଙ୍କା 8. $6\frac{2}{3}\%$ 9. 600ଟଙ୍କା
 ଓ 400ଟଙ୍କା 10. 500ଟଙ୍କା 11. 405ଟଙ୍କା 12. 500 ଟଙ୍କା 13. ପ୍ରଥମ ଦୋକାନରୁ କିଣିଲେ ଲାଭ ଜନକ ।
 14. 400ଟଙ୍କା, 15. 817.60 ଟଙ୍କା 16. 40% 17. 733.33 ଟଙ୍କା

ଅନୁଶୀଳନୀ - 8 (b)

1. (i) 36% (ii) 8ପଇସା (iii) 12.50% (iv) $\frac{25}{4}\%$; 2. 11,840ଟଙ୍କା 3. 8700ଟଙ୍କା 4. 1500ଟଙ୍କା
 5. 1620 ଟଙ୍କା 6. $\frac{P(9T+25)}{25}$ ଟଙ୍କା 7. 10ବର୍ଷ 8. 10,000ଟଙ୍କା 9. 10% 10. $6\frac{1}{4}\%$ 11. 2000ଟଙ୍କା
 12. $22\frac{1}{2}$ ବର୍ଷ 13. 16ବର୍ଷ, 14. 8% 15. 12,958ଟଙ୍କା, 16. 8%

ଅନୁଶୀଳନୀ - 8 (c)

1. ଟ.133.12ପ. 2. ଟ.1717.35 ପ. 3. 1655ଟ, 4. 1261ଟ 5. 6655ଟ 6. ଟ.36,659.70
 7. 2ବର୍ଷ 8. ଟ.567.45ପ 9. 3000ଟ 10. $3070\frac{5}{8}$ ଟ 11. ଟ 1125.21 ପ. 12. ଟ.103.37ପ
 13. ଟ.166116.80ପ 14. 22898 15. 38640 ଟ.

ଅନୁଶୀଳନୀ - 8 (d)

1. 10624 ଟଙ୍କା 2. 50 ଟଙ୍କା 3. 9000 ଟଙ୍କା 4. 147.40; 5. 129.01 6. 400 .20 7. 144.07
 8. 120.26 9. 122.65 10. 140

ଅନୁଶୀଳନୀ - 8 (e) 1. 100 ଟଙ୍କା; 2. 1000 ଟଙ୍କା; 3. ଦୁଇଥର 4. (a) 5ଟଙ୍କା; (b) କିଛି ସୁଧ ପାଇବ ନାହିଁ;

5. (i) Nil, (ii) ଟ.22.92, (iii) 22.92; 6. 127 ଟଙ୍କା; 7. 17.09 ଟଙ୍କା; 8. 144.58 ଟଙ୍କା;

9. 4.5%; 10. 293.00; 11. 42.43 ଟଙ୍କା; 12. 6%

ଅନୁଶୀଳନୀ - 9 (a) 1. x (କମଳାର ସଂଖ୍ୟା) : 8 4 18 10 13

y (କମଳାର ସଂଖ୍ୟା) : 18 14

2. (a) (i) 60 ଟଙ୍କା (ii) 5 ଟି, (b) (i) 700 ଟଙ୍କା (ii) 6 ଦିନ, 3. 15 ଟି 4. 225 ଟଙ୍କା, 5 ଟି, 5. 22.50 ଟଙ୍କା, 6
 କି.ଗ୍ରା., 6. 320 କି.ମି., 5 ଘଣ୍ଟା, 7. 7.50 ଟଙ୍କା; 8 ଟି, 8. 12 ଟଙ୍କା; 9. 5 ଲିଟର; 10. 4200 ଟଙ୍କା, 11. 720 ଟଙ୍କା, 12.
 150 ଟି, 108 ଟଙ୍କା

ଅନୁଶୀଳନୀ - 9 (b) 1. (i) ସଲଖ ଚଳନ (ii) ସଲଖ ଚଳନ ଅବଶିଷ୍ଟ ପ୍ରତିଲୋମୀ ଚଳନ

2. x : 10, 40; y : 8 4; k; 120 120 120 120 120 3. (i) ପ୍ରତିଲୋମୀ, (ii)
 ସଲଖ, (iii) ସଲଖ, (iv) ପ୍ରତିଲୋମୀ, 4. 1 ବ.ମି., 5. 12 ଜଣ, 6. 6 ଟି, 7. 12 ଦିନ, 8. 25 ଟି, 9. 1 : 2 : 3,
 10. 6 ଦିନ, 11. 20 ଏକର, 12. 10 ମିନିଟ୍, 2 କି.ମି.

ଅନୁଶୀଳନୀ - 9 (c) 1. 6 ଦିନ, 2. 20 ଦିନ, 3. 20 ଦିନ, 4. 10 ଦିନ, 5. 324 ମିଟର, 6. 8 ଜଣ, 7. 10 ଘଣ୍ଟା, 8.
 27 ଦିନ

ଅନୁଶୀଳନୀ - 10 (a) 4. (a) 11 am. (b) 5 am (c) 11 am (d) 20°C (e) 40°C 5. (a) 40, 45, (b)
 50-55, (c) 16, (d) 40-45, (e) 5; 7. (a) 120 (b) 90 (c) 5.4 (d) 90; 8. (a)
 270, 360, (b) 135, 135, (c) 180, (d) 135

ଅନୁଶୀଳନୀ - 10 (b) 1. x-ଅକ୍ଷ (b) y-ଅକ୍ଷ (c) (0,0) (d) y-ଅକ୍ଷ, (e) x-ଅକ୍ଷ (f) 0, (g) 0, (h) 3 (i) 1 (j)
 C(3,0)